



# Giacomo Castagnetti

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Master student at ETHZ, building software tools to solve biological problems.

## Work experience

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### ModLab ETH Zurich - De-novo Drug Design Internship

Nov 2025 – Mar 2026

- Setting up a precise **validation framework** that highlighted a failure mode of a drug design model.
- **Improved** training time by **20%** and **implemented** multiple successful **solutions** to the problem.
- Provided **visualizations** and reports for non-technical people on model limitations and next steps.

### Biozentrum - Bioinformatics Internship

Jul 2023 – Sep 2023

- Selected to the Biozentrum Research Summer program, in [Richard Neher's](#) lab.
- Developed a **novel tool** for tracking recombination in bulk ONT sequencing data (10x faster than the traditional experimental pipeline) ([GitHub](#)).
- Tracked the evolution and spreading of West Nile Virus (**Nextstrain**) ([GitHub](#)).
- Conducted a wet-lab directed evolution experiment.

## Education

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### Master in Computational Biology and Bioinformatics ([link](#))

ETH Zurich, Switzerland

Combining rigorous training in computer science, mathematics, and biology.

**GPA: 5.2** | 2024 - 2026

- Relevant coursework: Machine Learning, LLMs, Imaging of Organoids, Mathematical Modelling, Big Data

### Bachelor in Genomics ([link](#)) - graduation cum laude and honorable mention

University of Bologna, Italy

Highly interdisciplinary bachelor which gave me theoretical foundations in statistics, programming and molecular biology; together with practical bioinformatics skills.

**GPA: 6** | 2021 - 2024

## Publications

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Automated bacteriophage evolution studies with the Aionostat. [doi.org/10.1101/2025.04.16.649189](https://doi.org/10.1101/2025.04.16.649189)

April 2025

## Projects

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### Phagos Hackaton

Paris, France | May 2025

Used large protein and nucleotide foundation models to classify bacteriophage lifestyle.

### President of student association CCBB

Sep 2025 - Sep 2026

## Skills

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Programming Languages: Python (PyTorch, PyG, Pandas, Numpy, Snakemake), R (shiny), bash, C, C++, SQL, NoSQL

Technical: AI (ML, DL, LLMs, GNNs, LSTMs, VLMs, hugging face), Git, HPC, Scientific communication, Data Visualization, software packaging, experiment logging, AWS, cloud computing, representation learning, embeddings, data engineering, model evaluation, PK/PD.

Languages: English, Italian